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In [1]: # Program 2: Descriptive statistics on the Diamond dataset
import pandas as pd
from scipy import stats

FILE = r"D:\Datascience-MachineLearning\Datasets\diamonds.csv" # <-- change to your file path

df = pd.read_csv(FILE)
num = df.select_dtypes(include="number") # only numeric columns

for col in num.columns:
    x = num[col].dropna()
    print("\n===", col, "===")
    print("Mean          :", x.mean())
    print("Median         :", x.median())
    print("Mode           :", x.mode()[0])
    print("Std deviation    :", x.std())
    print("Variance         :", x.var())
    print("2nd moment       :", stats.moment(x, moment=2))
    print("3rd moment       :", stats.moment(x, moment=3))
    print("4th moment       :", stats.moment(x, moment=4))
    print("Skewness         :", stats.skew(x))
    print("Kurtosis         :", stats.kurtosis(x))
```

=== carat ===

Mean : 0.7979397478680014
Median : 0.7
Mode : 0.3
Std deviation : 0.4740112444054184
Variance : 0.22468665982277333
2nd moment : 0.22468249433037768
3rd moment : 0.11892064641252363
4th moment : 0.21487291967322694
Skewness : 1.1166148681277797
Kurtosis : 1.2564076184374766

=== depth ===

Mean : 61.749404894327036
Median : 61.8
Mode : 62.0
Std deviation : 1.432621318833661
Variance : 2.0524038431766978
2nd moment : 2.0523657934206136
3rd moment : -0.2419571081336167
4th moment : 36.809499412227936
Skewness : -0.08229173779627727
Kurtosis : 5.738771345086844

=== table ===

Mean : 57.45718390804598
Median : 57.0
Mode : 56.0
Std deviation : 2.2344905628213225
Variance : 4.9929480753375515
2nd moment : 4.992855510486321
3rd moment : 8.890229740837091
4th moment : 144.62295743573333
Skewness : 0.7968736878796518
Kurtosis : 2.801485914361578

=== price ===

Mean : 3932.799721913237
Median : 2401.0
Mode : 605
Std deviation : 3989.439738146379

Variance : 15915629.42430145
2nd moment : 15915334.362576861
3rd moment : 102753394353.22559
4th moment : 1311419991232699.0
Skewness : 1.6183502776053016
Kurtosis : 2.177382669056634

=== x ===

Mean : 5.731157211716722
Median : 5.7
Mode : 4.37
Std deviation : 1.1217607467924928
Variance : 1.258347173044451
2nd moment : 1.2583238443983065
3rd moment : 0.5344963402912307
4th moment : 3.7712687336783266
Skewness : 0.3786658120772097
Kurtosis : -0.6182146042773282

=== y ===

Mean : 5.734525954764553
Median : 5.71
Mode : 4.34
Std deviation : 1.1421346741235552
Variance : 1.3044716138353198
2nd moment : 1.304447430082746
3rd moment : 3.6264201119921697
4th moment : 160.29932193137375
Skewness : 2.4340990250113648
Kurtosis : 91.20599095863467

=== z ===

Mean : 3.5387337782721544
Median : 3.53
Mode : 2.7
Std deviation : 0.7056988469499941
Variance : 0.4980108625865512
2nd moment : 0.4980016299046345
3rd moment : 0.5350190913880986
4th moment : 12.420653215317506

Skewness : 1.5223802221853722
Kurtosis : 47.08214348390816

In []: